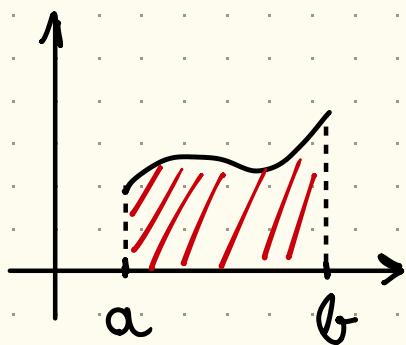
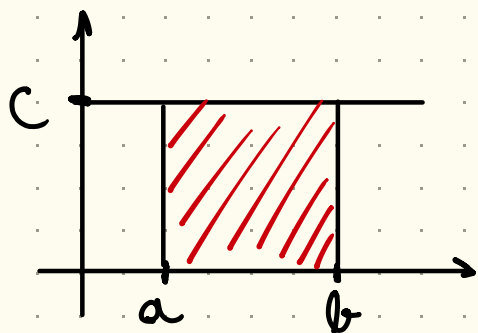


INTEGRALI DEFINITI



. STRUMENTO FONDAMENTALE

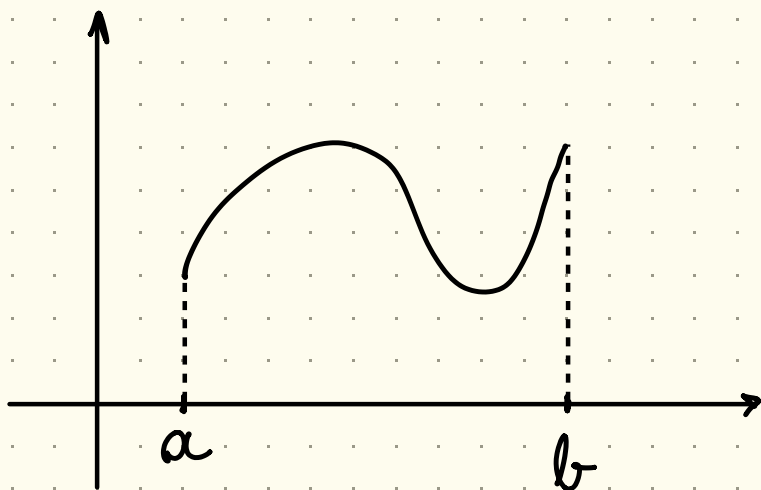


. SOMME INFERIORI

SIA $f: [a, b] \rightarrow \mathbb{R}$ LIMITATA

$$\exists m \leq M \text{ T.C.}$$

$$m \leq f(x) \leq M \quad \forall x \in [a, b]$$



$$S_0 = (b-a) \cdot \inf f(x)$$

$$S_1 = \frac{b-a}{2} \cdot \inf_{x \in [a, \frac{a+b}{2}]} f(x)$$

→ S_k É CRESCENTE (DEBOLMENTE)

INTUITIVAMENTE:

$$\lim_{k \rightarrow +\infty} S_k = \text{AREA}$$

. SOMME SUPERIORI

$$S_0, S_1, S_2, \dots, S_k \quad \begin{matrix} \text{COME PRIMA} \\ \text{INF} \leadsto \text{SUP} \end{matrix}$$

→ S_k É DECRESCENTE

$$\left. \begin{array}{l} \exists \lim_k S_k = S \\ \exists \lim_k S_k = S \end{array} \right\} \in [m(b-a), M(b-a)]$$

• DEFINIZIONE

SIA $f: [a, b] \rightarrow \mathbb{R}$ LIMITATA

f SI DICE INTEGRABILE (SECONDO RIEMANN)
SE:

$$s = S$$

IN TAL CASO DEFINIAMO L'AREA:

$$\int_a^b f(x) dx = s = S$$